

L. THARINDU NIRMAL WICKREMASINGHE

[webpage](#)tharindu.ncr@gmail.com[Profiles](#)[Documentation](#)

Research Interests: Generative AI, Computational Imaging, Image/Video restoration, Machine learning. Leveraging vision and language models for physically consistent restoration of images/video.

EDUCATION

PhD student: Purdue University

GPA: 4.00/4.00

2024-present

Intelligent Imaging Lab, Electrical & Computer Engineering

National Honor Society inductee- Purdue chapter of Phi Kappa Phi.

B.Sc.(Eng.): University of Moratuwa, Sri Lanka

GPA: 4.16/4.20(First Class Hons.)

2018-2023

Dean's List in all Semesters. Ranked 1/118 (Gold medal) in the graduating class for Electronics and Telecommunications Engineering. Ranked 1/980 (Gold medal) from the Faculty of Engineering.

EXPERIENCE

Graduate TA/RA- Purdue University, USA

Aug 2024 - present

Junior Instructor- University of Moratuwa, Sri Lanka

Dec 2023 - Aug 2024

Technical Consultant- IQ labs, Sri Lanka

Fall 2023

Remote Undergraduate Researcher- Centre for Advanced Imaging, Harvard University

2022-2023

Research Intern-School of Computing, University of Sydney, Australia

2021-2022

GRADUATE RESEARCH AND PUBLICATIONS

FlowSteer: Text-to-Video Generation via Neural Newtonian Dynamics

Apr 2025 - Nov 2025

Physics-Consistent and Controllable Text-to-Video Generation

CVPR'26 Project page

- Pioneered a zero-shot framework for flow-models across multiple image restoration tasks. Produced images with high pixel-level fidelity, while also achieving high perceptual quality.
- **Tharindu Wickremasinghe**, Chenyang Qi, Harshana Weligampola, Zhengzhong Tu, Stanley H. Chan. FlowSteer: Conditioning Flow Field for Consistent Image Restoration(arxiv)

SeeU: Seeing the Unseen World via 4D Dynamics-aware Generation

Aug 2025 - Dec 2025

Physically Consistent Video Restoration

CVPR'26 Project page

- Pioneered a 2D $\rightarrow$ 4D $\rightarrow$ 2D learning pipeline that lifts 2D inputs, models 4D dynamics, and projects back to produce coherent 2D results—introducing a new information flow and training scheme. Guided the generation of the 2D video within a realistic spatial and temporal context.
- Yu Yuan, **Tharindu Wickremasinghe**, Zeeshan Nadir, Xijun Wang, Yiheng Chi, Stanley H. Chan. SeeU: Seeing the Unseen World via 4D Dynamics-aware Generation (arxiv)

NewtonGen: Text-to-Video Generation via Neural Newtonian Dynamics

May 2025 - Oct 2025

Physics-Consistent and Controllable Text-to-Video Generation

ICLR'26 Project page

- Developed a physics-consistent and controllable text-to-video framework that explicitly incorporates learnable dynamics. Innovated novel metrics to monitor the Newtonian dynamics of the subject with different motions.
- Yu Yuan, Xijun Wang, **Tharindu Wickremasinghe**, Zeeshan Nadir, Bole Ma, Stanley H. Chan. NewtonGen: Physics-Consistent and Controllable Text-to-Video Generation via Neural Newtonian Dynamics. (arxiv)

OTHER RESEARCH PROJECTS

IEEE-IEEE ME-UYR selected young researcher program

Jan - Fall 2024

Affine Subspace Models and Clustering for Patch-Based Image Denoising

IEEE ACSSC 2025

- Research subspace models of image patches under the mentorship of prof. Marco Duarte. Introduced Affine Subspace Clustering to improve the image denoising baseline by 15%.

- **T. Wickremasinghe**, Marco F. Duarte; *Affine Subspace Models and Clustering for Patch-Based Image Denoising*. Presented our work at ICASSP 2025. Full paper accepted at Alisomar Conference on Circuits, Signals and Systems (ACSSC) 2025.

Masked Optimisation with Selective Attention for Image Reconstruction

Nov 2022 - 2024

Undergraduate research, collaboration with Wadduwage lab, Harvard university

Preprint 

- **T. Wickremasinghe**, A.Niwarthana, P. Somarathne, A. Thieshanthan, B. Weissbourd, C. Edussooriya, D. Wadduwage; *Lost in Translation: The Critical Role of Attention in Adaptive Image Reconstruction*, OSA Optica Imaging Congress, Cambridge, MA, 2023. (Extended abstract, Conference presentation)

Hierarchical Graph Neural Networks

May - Dec 2021

Self Initiated Undergraduate Research

ICPR 2022 publication 

- A. Thieshanthan, A. Niwarthana, P. Somarathne, **T. Wickremasinghe** and R. Rodrigo, "HPGNN: Using Hierarchical Graph Neural Networks for Outdoor Point Cloud Processing," in 2022 26th International Conference on Pattern Recognition (ICPR), Montreal, QC, Canada, 2022 pp. 2700-2706.

Real-time Configuration of Intelligent Reflecting Surfaces

Feb - June 2021

Research competition- ICASSP 2021

Video 

Report 

Announcement 

- IEEE Signals Processing Cup 2021. Our solution was **awarded 1st** place, invited to present at **IEEE ICASSP 2021-Toronto, Canada**. Supervised by Dr. P. Dharmawansa

Real-time object detection

Fall 2020 - 2021

Undergraduate research

IV Symposium 2022 publication 

- O. Jayasinghe, S. Hemachandra, D.Anhettigama, S. Kariyawasam, **T. Wickremasinghe**, C. Ekanayaka, R. Rodrigo, P.Jayasekara, "Towards Real-time Traffic Sign and Traffic Light Detection on Embedded Systems," 2022 IEEE Intelligent Vehicles Symposium (IV), Aachen, Germany, 2022, pp. 723-728.

Smart Breadboard for remote lab experiments

October 2020 - Feb 2021

Self supervised Research and Innovation

Video and Report- 

- Remote Access Breadboard to assist electronics laboratory work during lockdowns in the Pandemic. **1st place** at IEEE **PES** national design competition, Selected from IEE **CAS** student design competition 2020-2021, to represent Sri Lanka in the IEEE region 10 (Australia-Asia-Pacific).

AWARDS, SCHOLARSHIPS, AND INVITED PROGRAMMES

Gold medal for the most outstanding graduand in the faculty of Engineering, University of Moratuwa, Sri Lanka. For overall excellence in academics, extra curriculars, leadership, and civic service. *2023*

UNESCO Gold Medal for the best academic performance among all faculties, University of Moratuwa, Sri Lanka. Awarded at the general convocation of the university. *December 2023*

Pre-doctoral invitee, Cornell-Maryland Max Planck Research School (**CMMRS**), Germany. *August 2023*

SGI 2023 Selected to audit the workshops of the Summer Geometry Initiative at MIT- MA,USA. *August 2023*

COMPETITIONS AND HACKATHONS

Runners up	Datathon 2023 - Inter University Data science competition <i>An anomaly detection of suspicious activities on a website</i> 	<i>SLIIT</i>
Champions	Datathon 2022 - Inter University Data science competition <i>A recommending system for a telecommunications company using Next Best Action.</i> 	<i>SLIIT</i>
Champions	Data storm 2021- All island open Data science and analytics hackathon <i>Predict customer status and recommend the most suitable product.</i> 	<i>RCUOM</i>

Champions	Belt and Road Electricity simulation competition 2020 <i>Economical and energy efficient power distribution</i> 	Shanghai University of Power
Champions	Decrypt 2019- Product Innovation Ideathon <i>Heart disease and emergency alerting app for drivers</i> 	University of Moratuwa

TEACHING AND PEER REVIEW EXPERIENCE

Graduate Teaching Assistant: Purdue University-USA
ECE301:Signals and Systems.(Fall 2024)

Teaching Assistant, Junior Lecturer: University of Moratuwa-Sri Lanka
EN4553:Machine Vision, EN3551:Digital Signal Processing, EN3160: Machine Vision.(Fall 2023)
EN2412:Electronic Control Systems, EN2091:Laboratory practise. (Fall 2022)

Peer reviewer and reviewing assistance
ECCV 2026, ICML 2026, ICASSP 2025,2026, ISCAS 2024, CVPR 2023 (reviewing assistance), ACM ISS 2022, ICPR 2022.

TECHNICAL SKILLS

Languages and frameworks Python(Numpy, Pandas, Sklearn, Pytorch, Tensorflow), MATLAB, C++, HTML, Markdown, Latex.

Software and technologies Diffusers, Transformers, Huggingface, CUDA, OpenCV, Solidworks, Altium, Multisim, LTSpice, Webots, Slurm, Git, Bash.

MOOCs: Mathematics for Machine Learning- Imperial College London- (*Coursera, August 2020*), Neural Networks and Deep Learning- Deeplearning.AI (*Coursera, July 2020*), Computational Thinking and Datascience-MIT 6.00.2 (*EdX, June 2020*), Machine Learning- Stanford University (*Coursera, May 2020*).

VOLUNTEERING, LEADERSHIP AND EXTRA-CURRICULARS

Lecturing, Instructing and Community service

- Co-founder-Danuma Yathra Foundation, volunteering in rural schools of Sri Lanka, teaching highschool Mathematics and Science in schools with severe shortage of STEM teachers 2018-2020.
- Volunteer lecturer and instructor at astronomical society events 2018-2022.

Featured Talks, Presentations, and Articles

- Moderator on the panel “*Future of AI in Engineering*” at EXMO-2023 Flagship Public Engineering Exhibition of the University of Moratuwa, July 2023.
- Poster Presentation at the International Conference on Communicating Astronomy with the Public (CAP) 2022.
- Article by Emil Bjornson, Lucio Marcerano; *Configuring an Intelligent Reflecting Surface for Wireless Communications: Highlights from the 2021 IEEE Signal Processing Cup student competition*, IEEE Signal Processing Magazine,IEEE Signal Processing Magazine,Volume 39,Issue 1, January 2022, pp 126-131.

SELECTED AWARDS PRIOR TO UNIVERSITY

The **Most Outstanding Student** award in both Junior(2011) and Senior(2017) categories for the graduating class of Nalanda College, for overall excellence in academics and extra curriculars. 2011,2017

The **Best performance in Academics** award for Physical Sciences,

Overall Best in Academics in Nalanda College for the graduating class. 2017

2 Bronze medals - International Earth Science Olympiad (Individual and Field project), Brazil 2015

5 Bronze medals - International Junior Mathematics Olympiads, Indonesia 2009-2011

Gold medalist in the National Physics Olympiad *2017*, Bronze medalist in *2016*

2 Gold, 1 Silver, and 1 Bronze medal in the National Astronomy and Astro-Physics Olympiads in Junior (2012/13) and Senior categories (2014/15)

Class excellence with national rank **1** in the Royal Australian Chemical Institute Senior Quiz 2016